CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P1

Proposal and Requirements

Field	Your entry
Project title	Rep Rank
Team number and section	Team 3 and CS 3354.003
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0 Revision History and Contribution Record

Version	Date	Author	Summary of change
0.1	10/15/2026	David Wittler	Developed 1.1, 1.2, 1.3, 2.3, 2.4(slightly), 3.1, 3.2 (slightly), 3.3
0.1	10/16/2026	Veda Manchana	Revised 1.1, 1.2, 1.3, 2.3, 3.1, 1.4, developed 5
0.1	10/16/2026	Laxmi Komaragiri	3.2, 3.3, 4.4, 1.4
0.1	10/16/2026	Shreya Singaravel Sabapathi	Revised 1.4, Developed 2.2, 2.4, 2.5
0.1	10/16/2026	Anshika Anshika	3.4, 3.5 and 4.3
0.1	10/17/2026	Tanishq Seetala	2.1 and 2.6

0.1 Individual contribution record

Team member	What this person produced for P1	Evidence in the repository
David Wittler	2.6, 3.3, 4.2	Github PR #8 or check branch
Anshika Anshika	3.4, 3.5, 4.3 and final review of the document for submission.	GitHub PR #6 or check branch
Veda Manchana	1.2, 2.3, 3.1, 5	GitHub PR #4 GitHub PR #4 — Veda's P1 contribution documentation
Laxmi Komaragiri	3.2, 3.3, 4.4, 1.4, 2.6	-Technical Feasability XP Spike on SQLite and Flutter. Evidence is in main branch folder dev/spike/flutter_sqlite_demo folder - GitHub PR #3 or check branch dev/p1-documentation-laxmi
Shreya Singaravel Sabapathi	2.2, 2.4, 2.5	GitHub PR #5 - Shreya's P1 contribution documentation - https://github.com/AkshayaKomaragiri/software-engineering-project/pull/5
Tanishq Seetala	2.1 and 2.6	Github PR 7 or check branch Tanishq

1 Introduction

1.1 Purpose of this document and of the system

The purpose of this document is to present the initial proposal and software requirements for Rep Rank, a mobile application focused on personal fitness consistency and workout tracking. The document outlines the problem the application aims to address, identifies its intended users and stakeholders, and defines the system's scope. It also describes the functional and quality requirements, identifies the constraints that must be considered during development, and outlines the external services and dependencies the application will rely on. The document serves as a reference for the development team, course instructor, and stakeholders to evaluate the proposed system, and establish a shared understanding of its objectives.

Rep Rank is designed to help individuals who want to maintain consistent workout routines and monitor their fitness progress in a convenient way. The application aims to help users who may struggle to maintain a consistent workout routine, remember previous workouts, or keep track of their fitness goals. The application will allow users to search for exercises, log exercise details such as sets, repetitions, and weights, review previous workout records, and view exercise progress through line graphs. Users will also be able to create accounts and access their saved workout records across devices. The main goal of Rep Rank is to help users stay consistent with their workouts and track their personal progress over time.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Search for exercises by name using a search bar	Building our own cloud storage infrastructure, since we will use a third-party service
Log workouts by adding exercises and recording sets, repetitions, and weights	Building our own user authentication service, since we will use a third-party provider
View exercise progress over time through a line graph	Health diagnosis and medical advice, since Rep Rank is a fitness tracking app
Create an account and log in	Workout scheduling and calendars, since the initial version focuses on logging workouts rather than planning them.
View workout progress through a personal dashboard displaying exercise progress graphs and workout history.	Exercise rankings, competitive leaderboards, and weight-class competitions, since they introduce additional complexity and rely on user-reported exercise performance.
Access saved workout records across devices using the same account.	Wearable device integration requires additional development and testing.
Set weekly workout goals between one and seven sessions (COULD - optional).	

We will use third-party services for cloud storage and user authentication because we do not have the resources or infrastructure to build and maintain these services ourselves. Medical advice, workout scheduling, calendars, competitive leaderboards, and wearable integration are excluded to keep Rep Rank focused on workout tracking and personal progress visualization.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Rep (repetition)	One movement that completes an exercise, such as one squat, or one deadlift.
Set	A collection of reps performed together. A set of 8 reps for squat means the person did a complete squat movement 8 times. 3 sets of 8 means the person did this 3 times.
MAU (Monthly Active Users)	The number of unique users who use the application at least once during a calendar month.
Workout session	A completed workout that may include multiple exercises and sets.
Workout log	A record of a workout, including the exercises performed, sets, repetitions, and weights.
Weekly workout goal	Number of workouts a user aims to complete within a calendar week.
Progress dashboard	A screen that displays the user's workout history and exercise progress through line graphs.
Exercise performance	The weights, sets, and repetitions recorded for a particular exercise during previous workouts.
1RM (One-Rep Max)	Heaviest weight one can lift for a single, proper-form repetition of a specific exercise
Progress graph	A graph of One- Rep Max over time (One-Rep Max on y-axis and date on x-axis)

1.4 References

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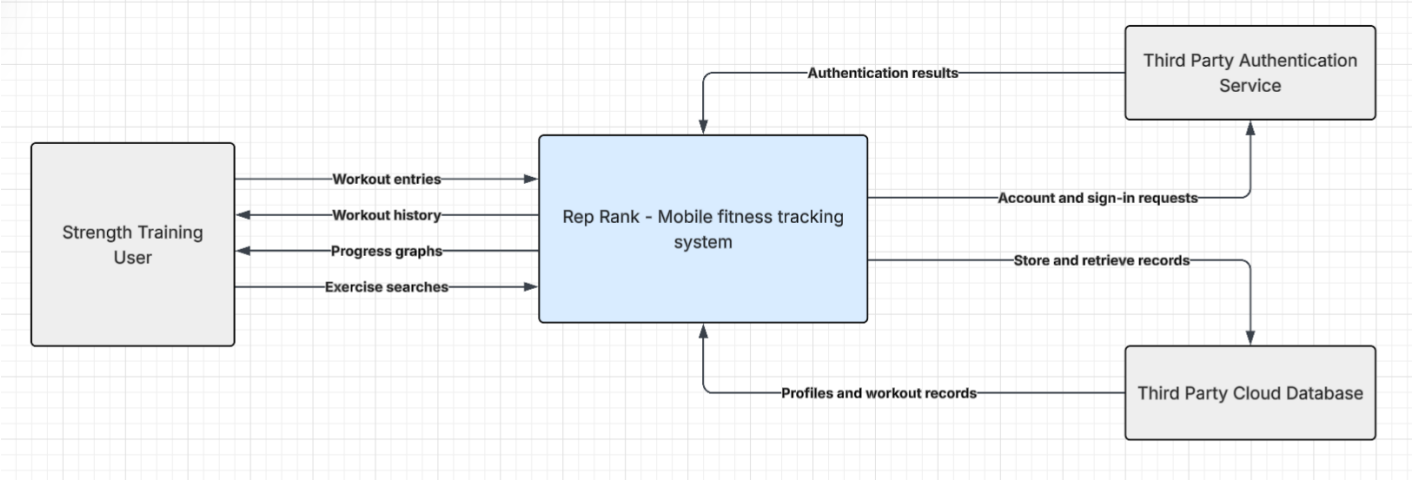
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2 Overall Description

2.1 Product perspective and context



Rep Rank is a mobile fitness tracking application that enables strength training users to search for exercises, record workouts, review previous records, and monitor progress through graphs. The system boundary includes the application interface, workout processing, progress calculations, local storage, and synchronization logic. Third-party authentication and cloud database services remain outside this boundary because external providers supply their infrastructure. Rep Rank connects to these services to authenticate users and save and retrieve account-specific records across devices. The team is responsible for configuring and testing these connections, protecting access to user records, and handling service failures. Any external exercise catalogue service selected later will also be shown outside the system boundary.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Strength Training Users	Primary Users of Rep Rank	They need a convenient way to record their exercises, review previous workout sessions, and monitor changes in their exercise performance over time.	This is an assumption based on personal gym experiences and observations. No formal interviews were documented in this process.

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Development Team	System designers, developers, and maintenance teams.	They require clearly defined requirements, manageable development scope, and maintainable application.	This was identified through the team's project planning and discussions regarding development responsibilities.
Third-party cloud and authentication providers	External services used for account authentication and data storage.	They require compatible integration, correct usage of services, and compliance with service requirements.	This was identified from the decision to outsource authentication and cloud storage in Section 1.2. No formal provider consultation documented.
App distribution platforms	External stakeholders responsible for hosting and distributing the application.	They require applications to meet the applicable platform requirements for installation and documentation.	This is an assumption based on the decision to support iOS and Android users. The specific platform requirements have not yet been formally reviewed.
Federal Trade Commission and relevant privacy authorities	Regulatory stakeholders are responsible for enforcing consumer protection and privacy requirements.	They require protection of user data privacy and truthful representation of the application's purpose and its capabilities.	This was identified through reviewing the FTC Mobile Health App Interactive Tool and Federal Trade Commission Act [R1, R8].

2.3 User characteristics

The primary users of Rep Rank are individuals who want to maintain consistent workout routines and track their personal fitness progress. These users include beginners who are developing regular exercise habits and more experienced individuals who want to organize their workout records and monitor their progress over time.

Beginner users are expected to have basic knowledge of mobile applications, but they may not be familiar with fitness terminology such as exercises, sets, reps (repetitions), one-rep max, and weights. They should not be expected to understand the application's internal data storage or authentication processes. Rep Rank should use clear labels, readable text, and simple navigation to help beginners search for exercises, record workout details, and understand their previous exercise performance.

Experienced users may already be familiar with strength-training terminology and regularly track their workout performance. These users may want to quickly review previous exercise records, compare weights across sessions, and view progress through line graphs on the progress dashboard. Since workout information is entered by the users, the accuracy of their records and progress graphs will depend on the information they enter.

Both user groups will primarily use iOS and Android mobile devices, often in gym environments. Users may record exercises during short breaks or while tired or distracted, so common actions like searching for exercises and entering sets, repetitions, and weights should be quick and easy to perform. Users may also have different accessibility needs, so the application should provide readable text, clear labels, and simple navigation.

2.4 Constraints

Rep Rank's design and development are subject to the following constraints.

1. Platform Requirements: The application must support iOS and Android operating systems. These platforms were specifically selected because they represent the majority of the Unites States mobile operating system market. Development and compatibility support for other operating systems are excluded from the initial release [R2].
2. Privacy and data handling: The application must protect any personal and fitness related information collected from users, including workout records and profile information. Personal workout information must not be publicly disclosed without appropriate user authorization except where disclosure is required by applicable law. This constraint limits how the workout records and personal information may be accessed, stored, or shared with third parties [R1].
3. Truthful health-related claims: Rep Rank must not make any false, misleading or unsupported claims regarding improvements in the users' physical health, strength or fitness. The application is intended for workout tracking and progress visualization rather than medical diagnosis or treatment. This constraint restricts the claims that may appear in the application's interface or promotional materials [R8].
4. External service integration: Based on the team's defined project scope and resource limitations, account authorization and cloud storage will rely on third party services rather than the infrastructure developed and operated by the team. This constraint rules out developing custom authentication and cloud storage infrastructure for the initial release (Section 1.2).

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	The selected third-party database and authentication providers are assumed to maintain appropriate security measures to protect user data against unauthorized access and data loss.	It results in user's personal information and workout records being exposed, compromised, or lost, undermining privacy and user trust in the application.
A2	The selected cloud service provider is assumed to maintain sufficient service availability, data integrity, and backup capabilities to support the applications' intended usage.	The users may not be able to save, retrieve, or synchronize workout records reliably across application sessions and devices.
A3	The targeted audiences are assumed to have access to a compatible iOS or Android device and possess familiarity with navigating mobile applications. Knowledge of specific exercise terminology is not assumed.	The prospective users without compatible devices or basic mobile navigation skills may struggle to access the application or use its workout tracking features effectively.
A4	The users are assumed to accurately record their workout performance, including the weights lifted and number of sets and repetitions completed.	Incorrect or inconsistent workout entries may result in inaccurate workout histories and misleading progress graphs, reducing the reliability of the application's progress tracking functionality.
A5	A sufficiently comprehensive and usable exercise dataset is assumed to be available for the application's exercise search functionality.	The users may be unable to find exercises they perform, preventing the application from fully supporting its intended workout logging functionality.
A6	The selected third-party authentication and cloud services are assumed to provide necessary APIs or SDKs compatible with the application's iOS and Android development framework.	Account creation, authentication and workout synchronization may require a different service provider or changes to the application architecture, potentially delaying development.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible conditional on spike returning before 10/25/2026	Rep Rank's core requirements cover exercise search, workout logging, saved history, and progress graphs. Authentication and cloud storage infrastructure would be supplied by third-party providers, reducing the infrastructure the team must build. However, the team must demonstrate that its chosen framework and storage approach can save a workout, retrieve it after restarting the application, and display the recorded data. This technical spike (see dev/spike/flutter-sqlite-demo) will provide evidence specific to the team's ability to build the system.
Operational	Feasible, conditional on user validation	Sections 2.2 and 2.3 identify beginner and experienced users who need convenient workout logging, access to previous records, and progress tracking during short gym rest periods. These needs are based on personal experience and observations rather than formal interviews. The offline prototype planned in Section 4.2 will allow representative users to test exercise search, workout entry, and history retrieval. Their feedback will help confirm that the workflow fits their routines and identify improvements before cloud synchronization is introduced.
Schedule	Feasible	The project has been divided into manageable phases with defined deliverables, allowing progress to be monitored throughout development (4.2). Technical capabilities are still in question but a working fitness tracker will be available throughout the development after phase 3 (offline workout tracker with no unique user support).
Economic / resource	Feasible conditional on revenue scaling with MAU	The estimated operating costs of the application are relatively low during initial development and are expected to scale with the size of the user base. The most significant expense to consider is third party database support and transactional email services as the app scales. Supabase costs range from $0 during early development to $1,000 or more per month as monthly active users (MAU), database usage, storage, and bandwidth increase [R10]. Transactional email services also scale with usage, with an estimated cost of approximately $20 per 25,000 emails sent[R11]. Advertising revenue from the app is also expected to scale with the number of monthly active users. At 100,000 MAU, advertising revenue is expected to generate thousands per month, although actual revenue will vary significantly based on user engagement, geographic distribution, ad placement, and monetization strategy. Because the application's major operating expenses scale alongside user growth, while potential advertising revenue also increases with a larger and more active user base, the project is considered economically feasible. Initial operating costs should remain relatively low, with additional infrastructure expenses becoming significant primarily as the application gains users.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall allow users to record the weight, number of sets, and number of repetitions performed for each exercise during a workout.	MUST	Strength Training Users (2.2)	A tester enters an exercise with a specified weight, number of sets, and repetitions, saves the workout, and verifies that the recorded values match what was entered.
FR-02	The system shall save each exercise log to the user's workout history and retain previously recorded logs across application sessions.	MUST	Strength Training Users (2.2)	A tester saves an exercise log, closes and reopens the application, and verifies that the previously saved log is still available in workout history.
FR-03	When a user selects an exercise to log, the system shall display the user's previously recorded performance for that exercise.	MUST	Strength Training Users (2.2)	A tester records the same exercise with different weights and repetitions across multiple workouts and verifies that the most recent recorded values appear when selecting that exercise again.
FR-04	The system shall calculate an estimated one-repetition maximum (1RM) for a selected exercise using recorded weight and repetitions and display the results over time in a line graph.	MUST	Strength Training Users (2.2)	A tester records exercise weights and repetitions across three workout dates, opens the progress graph, and verifies that the estimated 1RM values are displayed for the corresponding dates and match the calculation method selected by the team.
FR-05	The system shall allow users to set a weekly workout goal between one and seven sessions.	COULD	Strength Training Users (2.2), R9	A tester sets a weekly goal of four workouts, saves it, and verifies that the displayed goal is four workouts per week.
FR-06	The system shall display a leaderboard ranking participating users by their highest recorded weight for a selected exercise.	WON’T	Strength Training Users (2.2)	A tester records different weights for the same exercise under two participating accounts and verifies that the leaderboard ranks the higher recorded weight first.
FR-07	The system shall allow users to create an account using an email address and a third-party authentication method.	MUST	Strength Training Users (2.2)	A tester registers using a valid email address through the third-party authentication provider and verifies that the account is successfully created.
FR-08	The system shall authenticate registered users before granting access to their private account information.	MUST	Strength Training Users (2.2)	A tester attempts to access private account information without signing in and verifies that access is denied.
FR-09	The system shall make a user's saved workout records available across devices when signed in to the same account.	SHOULD	Strength Training Users (2.2)	A tester saves a workout on one device, signs in to the same account on another device, and verifies that the saved workout is available.
FR-10	The system shall allow users to search for exercises by name when creating a workout log.	MUST	Strength Training Users (2.2)	A tester searches for an exercise by name and verifies that a matching exercise appears and can be selected.

3.2 Quality requirements

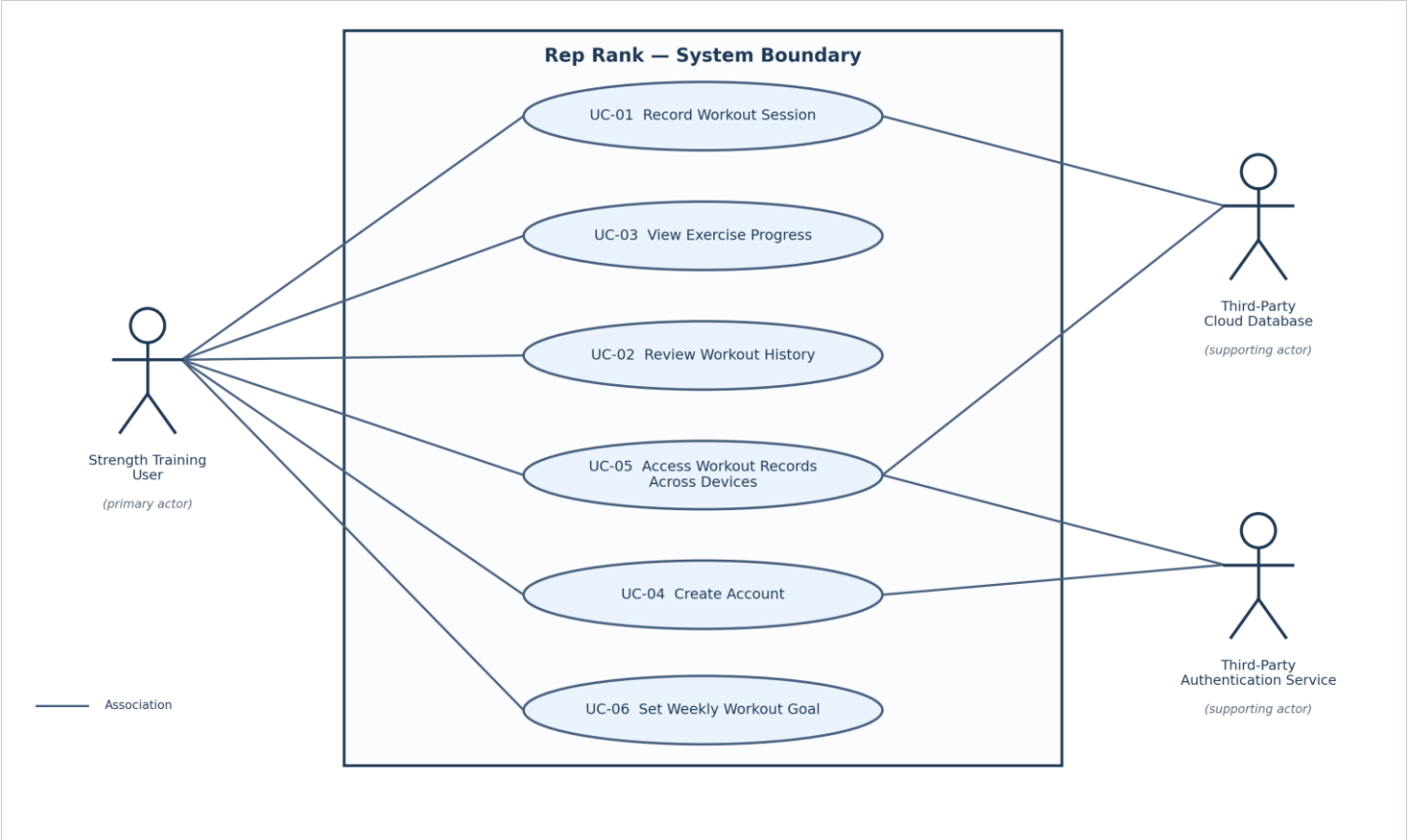
ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	Security – All authentication tokens issued by the third-party provider must be cryptographically verified using the provider's public keys or SDK before granting session access.	MUST	R3	Will be verified with automated unit test that intercepts an authenticated API request using a simulated valid token, invalid token, and expired token, confirming that unauthorized requests are successfully rejected.
QR-02	Performance – The mobile application must achieve a Cold Start time of less than 2.0 seconds under standard network conditions on supported iOS and Android device profiles.	SHOULD	R4	This will be verified using performance measuring tools like Xcode instruments for iOS and android studio profiler on the test devices. The elapsed time from app launch to the first interactive screen for 10 consecutive cold starts will be measured.
QR-03	Maintainability – The codebase should maintain a cyclomatic complexity of no more than 10 per method/function which will be measured through automated static code analysis tools (such as SonarQube) during code reviews and pull request integration.	MUST	R5, R4	This will be verified continuously with automated static code analysis gates such as SonarQube or ESLint/Pylint. It will be integrated into the GitHub Actions CI pipeline which will automatically fail any pull request with a cyclomatic complexity greater than 10.
QR-04	Accessibility – The mobile app must not restrict its view and operation to a single display orientation	COULD	R6	This will be verified manually by inspecting the UI layout, by rotating test devices between portrait and landscape modes across all primary app screens.
QR-05	Accessibility – Pointer and touch targets must be large enough to activate easily. This is 24 by 24 CSS pixels.	COULD	R7	This will be verified using automated accessibility tools, like Google Accessibility Scanner) along with manual UI inspect to confirm that all buttons and touch targets meet or exceed the minimum 24 by 24 CSS pixels threshold.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system needs to be compatible with IOS and android mobile operating systems	[R2]	More than 99% of all smartphones in the United States use these operating systems. This is essential for accessibility.
CON-02	The application shall maintain the confidentiality of users' personal and health-related information, including age, sex, weight, and workout history. Such information shall not be disclosed to third parties without the user's explicit consent, except where disclosure is required by applicable law.	[R1]	Legal compliance with legislation is mandatory.

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-03	The application shall not make false, misleading, or unsubstantiated claims regarding its effects on users' health, fitness, or physical performance.	[R8]	Legal compliance with legislation is mandatory

3.4 Initial use cases



Expanded use case — UC-01

Field	Content
Identifier and name	UC-01 - Record Workout Session
Primary actor	Strength Training User
Supporting actors	Third-party cloud database, when cloud storage is enabled
Preconditions — what must be true before it starts	The user has opened the application and has access to the workout logging feature.
Trigger — what starts it	The user chooses to record a workout session.
Main success scenario — numbered steps	1. The user opens the workout logging feature. 2. The system displays an exercise search bar. 3. The user searches for and selects an exercise. 4. The system displays the user's most recent recorded performance for that exercise, if available. 5. The user enters the weight, number of sets, and repetitions performed.

Field	Content
	6. The user saves the workout entry. 7. The system stores the exercise log in the user's workout history and, when cloud storage is enabled, saves it to the third-party cloud database. 8. The system confirms that the entry was saved successfully.
Alternative and exception flows	A1 (step 3): If the exercise search returns no matches, the system informs the user and allows another search. A2 (step 4): If the exercise has no previous records, the system allows the user to continue without displaying past performance. A3 (after step 8): If the user wants to record another exercise in the same workout session, the user repeats steps 3 to 8. E1 (step 6): If required workout information is missing or invalid, the system requests corrections before saving. E2 (step 7): If saving fails, the system informs the user that the workout was not saved and allows another attempt.
Postconditions — what is true when it succeeds	The workout entry is saved in the user's workout history and can be retrieved in a later application session.
Requirements realised	FR-01, FR-02, FR-03, FR-10

Expanded use case — UC-05

Field	Content
Identifier and name	UC-05 — Access Workout Records Across Devices
Primary actor	Strength Training User
Supporting actors	Third-party authentication service and third-party cloud database
Preconditions — what must be true before it starts	The user has an existing Rep Rank account. The application is installed on a second supported device that has network access.
Trigger — what starts it	The user opens Rep Rank on another device to access their saved workout records.
Main success scenario — numbered steps	1. The user opens the application on the second device. 2. The user selects the account sign-in option. 3. The system requests the user's account credentials. 4. The user submits their credentials. 5. The system requests authentication through the third-party authentication service. 6. The authentication service verifies the credentials. 7. The system retrieves the workout records associated with the user's unique account identifier from cloud storage. 8. The system displays the user's previously recorded workouts.
Alternative and exception flows	A1 (step 6): If the credentials are incorrect, the system displays an authentication error and allows the user to retry. A2 (step 8): If the account has no saved workout records, the system displays an empty workout history. E1 (step 5): If the authentication service is unavailable, the system informs the user that sign-in cannot be completed.

	E2 (step 7): If the cloud database is unavailable, the system informs the user that workout records cannot currently be retrieved. E3 (steps 1 to 6): If a user who has not yet signed in tries to open private workout records, the system denies access and requires successful authentication before displaying account-specific information.
Postconditions — what is true when it succeeds	The user is authenticated and can access workout records associated with their account on the second device.
Requirements realised	FR-08, FR-09

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realize it	If blank — why, and what you will do
FR-01	UC-01	
FR-02	UC-01, UC-02	
FR-03	UC-01	
FR-04	UC-03	
FR-05	UC-06	
FR-06		No use case is written for FR-06. It is a WON'T requirement, and competitive leaderboards are out of scope for the initial release (Section 1.2). The requirement stays in Section 3.1 for traceability. If it is brought back into scope, it will go through change control (Section 6) and a use case will be written at that point.
FR-07	UC-04	
FR-08	UC-05	
FR-09	UC-05	
FR-10	UC-01	

The matrix traces nine of the ten functional requirements to at least one of the six use cases. FR-06 is the only requirement with no use case, for the reason given in its row. UC-06 (Set Weekly Workout Goal) was added so that FR-05 has a use case. Every use case in the diagram in Section 3.4 traces to at least one functional requirement, so no use case is left without a requirement.

The matrix also shows three things. FR-02 is realized in two use cases: UC-01 saves the log, and UC-02 retrieves it. FR-08 and FR-09 trace only to UC-05, so UC-05 carries both sign-in and cross-device access. FR-07 traces to UC-04, and UC-04 and UC-05 are the two use cases that use the third-party authentication service.

4 Project Plan

4.1 Team roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Anshika Anshika	Backend Development and Cloud Integration Lead: Responsible for cloud database development, backend integration, and synchronization between local SQLite storage and Supabase.	Team discussion, followed by majority vote, if consensus cannot be reached.
David Wittler	System Analyst: Work with developers to ensure the UI, database, authentication, and synchronization components interact correctly. Define how user actions, such as recording or editing a workout, translate into database operations. Finalize the privacy policy based on actual data-handling practices, validate user access controls and account deletion processes, and verify that implemented functionality satisfies system requirements.	Team discussion, followed by majority vote, if consensus cannot be reached.
Laxmi Akshaya Komaragiri	Database Architecture & Security: Responsible for designing the Supabase cloud schema, setting up relational constraints, and configuring row-level security policies to ensure data isolation.	Team discussion, followed by majority vote, if consensus cannot be reached.
Shreya Singaravel Sabapathi	Local Database and Analytics Lead: Responsible for implementing account - specific SQLite storage, ensuring secure storage and retrieval of workout records, and developing workout history and progress visualization graphs to track users' exercise performance over time.	Team discussion, followed by majority vote, if consensus cannot be reached.
Tanishq Seetala	Frontend and UI/UX Lead: Responsible for designing and implementing the mobile application user interface which includes workout logging screens, exercise selection, progress graphs, dashboard and navigation. Ensure the interface is intuitive, responsive, and compatible with both iOS and Android platforms.	Team discussion, followed by majority vote, if consensus cannot be reached.

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Veda Manchana	Authentication and cross-platform testing lead. Responsible for implementing user registration, login, logout, and password recovery, as well as testing core application features on iOS and Android.	Team discussion, followed by majority vote, if consensus cannot be reached.

4.2 Plan to P2

Date	What will be finished	Owner
10/21/2026	Built database schema for workouts, exercises, and sets. Determine relations (tables), records (rows), primary keys (unique identifiers), foreign keys, functional dependencies (basically relationships), and multivalued dependencies-> (try to avoid these).	David Wittler
10/24/2026	Set up SQLite with a given schema. Set up and implement insertion, retrieval, updating, and deletion functions that will be used in the app. Deliverable: working local workout data storage	Laxmi Komaragiri
10/27/2026	User Interface: build the screens for selecting exercises, recording sets and repetition, viewing previous workouts, and setting up logs. Deliverable: A usable offline workout tracker, the look of the UI can be sub-par	Tanishq Seetala
10/29/2026	Account Creation: Integrate Supabase Authentication, implement registration, login, logout, and account recovery Deliverable: account creation, login, logout, account recovery and account deletion all usable	Veda Manchana
11/01/2026	Establish account-specific local storage (SQLite) and prevent one account from accessing another account's records. Deliverable: Secure account management, different accounts supported on one device	Shreya Singaravel Sabapathi
11/03/2026	Cloud Database and synchronization: Create the corresponding Supabase tables (similar schema used for SQLite). Implement uploading local change and downloading cloud updates. (Changes in SQLite database are synched with Supabase and changes in Supabase are synched to SQLite database) Deliverable: Cross-device workout history, and offline support	Anshika Anshika
11/06/2026	Implement progress graphs and history of previous sessions using the accounts associated data. Start cleaning up UI Deliverable: Progress graph feature and workout history feature.	Shreya Singaravel Sabapathi
After P2, if time permits	Implement a weekly workout goal feature (COULD), further clean up UI Deliverable: A screen dedicated to setting up a weekly workout goal	Laxmi Komaragiri
11/08/2026	Security, Privacy, and release preparation: Finalize privacy policy based on actual data handling. Validate access flows (users can only modify their own data). Deletion flows (when a user deletes their accounts), backups, accessibility, and app-store requirements Deliverable: Rep Rank version 1.0	David Wittler

Note: This plan identifies proposed development milestones through November 8, 2026. The team will adjust the deliverables as necessary to align with the confirmed P2 requirements.

4.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	Team members may miss deadlines for assigned tasks	Moderate	High	David Wittler: By 10/19/2026, assign a deadline to every GitHub issue, review task progress twice per week, and reassign overdue tasks at the next team meeting.
R2	Third-party authentication or cloud database integration may fail or take longer than expected	Moderate	High	Anshika Anshika: By 10/28/2026, build an early integration prototype that creates a test account, signs in, saves a workout record, and retrieves it from the cloud database. Document integration errors and create GitHub issues for unresolved problems.
R3	Workout records may be lost, incorrectly saved, or associated with the wrong user	Moderate	High	Laxmi Akshaya Komaragiri: By 11/05/2026, write and run tests for saving and retrieving workout records, including tests that confirm records stay associated with the correct user account.
R4	The application may behave differently across iOS and Android devices	Moderate	Moderate	Veda Manchana: By 11/07/2026, test the main application features using both iOS and Android development environments and record compatibility issues in GitHub.
R5	Additional features or changing requirements may cause the project to exceed its planned scope	High	High	Shreya Singaravel Sabapathi: Beginning 10/19/2026, compare proposed features against the approved scope in Section 1.2. Document requested changes in GitHub and obtain team approval before adding new features to the development plan.

4.4 Repository and tooling

Repository URL: https://github.com/AkshayaKomaragiri/software-engineering-project

Branch Conventions and Rules

- Branching
  - Each full feature or bug fix will be built on a separate branch. The naming convention for branches must be one of either bugfix/<featurename> /<fix name>, update/<feature name>/<update type>, or dev/<feature name>.
  - There should not be more than one dev type branch per feature.
  - No more than two people should be working on the dev branch for a feature.
- Commits
  - Each commit must be prefixed by:
    - feat: - new feature
    - fix: - bug fix

    - docs: - adding/updating documentation
    - refactor: - cleaning up code
    - test: - adding/updating tests
- Will use Github's kanban board to keep track of tasks. The issues are either in backlog, ready, in progress, in review, or done
- Before a pull request can be merged, at least one other member of the group must review it. The person that submitted the pull request is responsible for tagging who they would like to review the pull request. The reviewer must review the request within 3 to 4 days of being tagged. If the reviewer suggests any changes, the changes need to be completed within one week. Any cases where this is not possible, the group member must communicate with the rest of the team in the designated communication channel.
- All team members have access to the repository and can make changes to it.

4.5 Declaration of AI tool use

Tool	What it was used for	Who used it	How the output was checked
ChatGPT	Extracting functional requirements from statements about the apps intended functions.	David Wittler	Just a thorough read over the functional requirements to make sure they aligned with my intended functions.
ChatGPT	Rewriting my assumptions in a more precise and professional manner	David Wittler	Reading over the output, making sure my ideas were conveyed
ChatGPT	Reviewing and refining Sections 1.1, 1.2, 1.3, 2.3, and 3.1 for clarity, consistency, and alignment with the project scope.	Veda Manchana	Manually reviewed the suggestions, compared them with the project requirements and scope, and revised the document to ensure the final content reflected the team's decisions.
ChatGPT	Professionalizing the economic feasibility argument	David Wittler	Reading over the outputs making sure my ideas from the original feasibility argument were conveyed and nothing extra was added
Gemini	Understanding what the text of industry quality standards documentation as they were quite lengthy and wordy.	Laxmi Komaragiri	I used ai only as a summarizer, that would put the documentation language in a simpler language. Gemini told me which parts of the documentation that it used, and I went and reread it.
ChatGPT /Claude	Drafting and refining Sections 3.4, 3.5, and 4.3, reviewing UML use-case relationships, checking requirements traceability, and identifying inconsistencies across the P1 document.	Anshika Anshika	Compared to the suggestions against the professor's P1 template, the team's functional requirements in Section 3.1, the defined project scope, and the use-case diagram. Reviewed suggested changes for

Tool	What it was used for	Who used it	How the output was checked
			consistency and flagged unresolved design decisions for team discussion.
ChatGPT	Drafting and refining Sections 1.4, 2.2, 2.4, and 2.5, organizing references, identifying relevant stakeholders, defining project constraints, and reviewing assumptions and dependencies for clarity and consistency across the P1 document.	Shreya Singaravel Sabapathi	Reviewed the suggestions against the professor's P1 template, the team's project scope, and the existing functional and quality requirements. Also checked the wording, references, and consistency with other sections to ensure the final content accurately reflected the team's project decisions.
Chatgpt/Gemini	Drafting rephrasing and refining sections 2.1 and 2.6	Tanishq Seetala	Reviewed by a detailed reading and comparison against the lecture slide shows.

5 Open Questions

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q1	Which third-party authentication and cloud database services will Rep Rank use?	Anshika Anshika	10/25/2026	We assume suitable third-party services will be available and compatible with iOS and Android.
Q2	Will the team have enough time to implement weekly workout goals feature?	Veda Manchana	11/01/2026	We assume weekly workout goals will remain a COULD feature and will be implemented only if time and resources allow.
Q3	Which calculation method will Rep Rank use to estimate one-repetition maximum (1RM)?	David Wittler	10/20/2026	The team will select and document an estimation method to calculate the one repetition max before implementing and testing the progress graph.

6 Baseline Declaration

We confirm that the requirements in section 3 represent our agreed baseline as of the submission date, and that every team member has read this document in full.

Team member	Signature or typed name	Date
Anshika Anshika	Anshika	10/18/2026
Laxmi Komaragiri	Laxmi Komaragiri	10/18/2026
Tanishq Seetala	Tanishq Seetala	10/18/2026

Team member	Signature or typed name	Date
Shreya Singaravel Sabapathi	Shreya Singaravel Sabapathi	10/18/2026
Veda Manchana	Veda Manchana	10/18/2026
David Wittler	David Wittler	10/18/2026